Kuncheng Wu

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Research interests: Embodied AI and aerial–ground multi-robot coordination; VLM-based planning and closed-loop control; world models and world-action models (WAMs); spatial memory and active perception; reinforcement learning for tool-using agents.

Research experience spans VLM training and evaluation, aerial–ground robot coordination, autonomous navigation, and memory-based reasoning in dynamic environments.

EDUCATION

University of California, Berkeley | Aug 2025 – May 2026

M.S. in Civil and Environmental Engineering; Transportation Systems. GPA: 3.74/4.00.

University of California, Berkeley | Aug 2024 – May 2025

Visiting Student, Civil and Environmental Engineering. GPA: 3.85/4.00.

Tongji University | Sep 2021 – Jul 2025

B.Eng. in Transportation Engineering. GPA: 4.28/5.00.

PUBLICATIONS — ACCEPTED

1. Qing Lyu*, Jinglei Zeng*, **Kuncheng Wu***, and Jasenka Rakas†. **Dynamic Estimation of Airport Taxiway Hotspots Using Spatio-Temporal Traffic and Kinematic Risk Modeling.** Accident Analysis & Prevention, 2026. Accepted for publication. *Equal contribution (co-first authors); †corresponding author.

2. Meng Lu, Ligeng Zhu, Olivia Xiao, Yuchen Zhuang, Zihan Wang, **Kuncheng Wu**, Bangya Liu, Yu Wang, Charles Fleming, Wenqi Shi, and Xuan Wang. **VICO: Visual Environments Co-Evolving for Vision-Language Model Reasoning.** NeurIPS 2026, Main Track, Poster.

EMBODIED AI AND ROBOTICS RESEARCH

Air–Ground Multi-Robot Simulation and VLM Control | Jul 2026 – Present

Virginia Tech — collaboration between Prof. Xuan Wang's group and Prof. Nathan Lau's Systems Cognition Engineering Laboratory (SCEL).

- Built Isaac Sim and SimWorld workflows for Go2 navigation, drone inspection, and aerial-to-ground search-and-verification handoffs, connecting visual observations with VLM actions and environment feedback.
- Evaluated Qwen, InternVL, and DeepSeek policies on navigation, aerial search, and handoff tasks: success, collisions, stopping, localization error, and latency.
- Implemented and tested verification and coverage-memory variants for drone-to-Go2 handoffs, including moved-target and target-absent conditions to study failures caused by stale or unsupported reports.
- Public ESARBench UAV search-and-rescue benchmark (120 tasks): a claim–verify–commit agent reached 18.5% success vs 10.4% for APEX with the same VLM, at 93% report precision.

Robot Memory and World-Model Evidence for Embodied Reasoning | Ongoing

- Built a 760-query benchmark for deciding when to answer from robot memory, inspect again, or abstain, across four evidence states and three observation budgets.
- Developed a replayable dynamic-household pipeline with robot patrols, RGB/depth/segmentation capture, and visibility-derived memory; separated structured-memory reasoning from visual perception errors.
- Compared structured memory, last-seen images, and MindJourney-generated views in a controlled diagnostic of reasoning with incomplete or outdated observations.

Autonomous Robot Navigation — Unitree Go2 EDU | May 2025 – Mar 2026

- Integrated LiDAR–inertial SLAM, persistent point-cloud maps, and Open3D ICP localization on an Orin NX.
- Bridged ROS 2/ROS 1 and debugged memory limits, IMU calibration, and remote operation on the robot.

MULTIMODAL MODELS AND LEARNING

Research Intern — Virginia Tech | 2026 – Present

Xuan Wang's research group, Department of Computer Science.

VICO: Visual Environments Co-Evolving for VLM Reasoning | NeurIPS 2026

- Reproduced and ran baseline methods for comparative evaluation of vision-language model reasoning.
- Contributed to model training experiments within the actor–environment co-evolution framework.

Tool-Augmented Video Reasoning and VLM Post-Training | Ongoing

- Built a surgical-video agent combining temporal recognition, event timelines, targeted frame inspection, and instrument-focused tools; fine-tuned a 9B VLM with LoRA on 2,955 tool-use trajectories.
- Evaluated on 3,877 internally constructed questions across CholecT50, HeiChole, and MultiBypass; achieved 69.7%, 64.9%, and 55.1% accuracy, improving over matched answer-only SFT by 4.5, 10.6, and 7.4 percentage points.
- Post-trained with GRPO (dynamic sampling) on up to 12 A100s across preemptible Slurm jobs: 69.7% → 71.1% in-domain (McNemar $p = .011$); 74.4% with added evidence tools, on par with DeepSeek-V4.1-Flash using the same tools (73.3%).
- Used five-fold cross-fitting and image-removal/misalignment ablations to study tool errors, visual evidence use, and transfer across hospitals and procedures.

SPATIAL REPRESENTATIONS AND EARLIER RESEARCH

Neural Scene Representations for World Modeling | Jan 2026 – Present

- Developing NeRF and 3D Gaussian Splatting pipelines from video to investigate persistent 3D representations for spatial reasoning and navigation.
- Studying reconstruction quality, inference cost, and the use of reconstructed scenes for simulation.

Graph Neural Networks for Airport Site Selection — Tongji University | Jan 2024 – Jul 2025

Undergraduate thesis; advisor: Prof. Shifu Liu.

- Built a heterogeneous graph from GIS and socioeconomic data; trained a graph attention network for four-dimensional feasibility estimation and applied Pareto optimization to compare site alternatives.

Shared Micromobility Optimization — UC Berkeley | Began Sep 2025

Research Assistant; advisor: Prof. Manxi Wu.

- Analyzed stochastic MILP, rider incentives, and queueing models for vehicle rebalancing.

INDUSTRY EXPERIENCE

Transport Planning Intern — Tongji Architectural Design & Research Institute | Began Jun 2023

- Analyzed traffic data and supported infrastructure drafting for the Shantou Railway Station transportation hub.

HONORS AND AWARDS

MCM Meritorious Winner (2024); National College Students Mathematical Modeling Contest, Shanghai Third Prize (2023); “Several Dimensional Cup” International Mathematical Modeling Challenge, Second Prize (2022).

TECHNICAL SKILLS

Models and learning: PyTorch, VLM evaluation, LoRA/SFT, GRPO, multi-node training, tool-use trajectories, graph neural networks, scikit-learn.

Robotics and simulation: Isaac Sim, SimWorld, Unreal Engine/UnrealCV, ROS 2/ROS 1, Go2, Orin NX, Open3D, SLAM.

Spatial representations: NeRF, 3D Gaussian Splatting, point clouds, scene graphs, GIS.

Programming and evaluation: Python, R, MATLAB, C++; Linux, vLLM, dataset auditing, paired statistics.